

The History of Computing

- **Impact of Innovation:** The rise of technological change and innovation over recent decades has brought immense significance to society. Computers, mobile devices, the Internet, the World Wide Web (WWW), and social media are among the advancements that have improved the standard of living.
- **Computing Definition:** Computing is any activity that utilizes computers to manage, process, and communicate critical information that is an integral part of modern industrial technology.

Early Civilizations

- **Foundation:** The naive view of calculation must be overcome to utilize the power of computing. The contributions of early civilizations form the foundation of modern computing.

The Babylonians

- **Timeframe & Location:** Thrived in Mesopotamia (Iraq) from about 2000 BC until about 500 BC.
- **Mathematical Texts:** Clay cuneiform tablets deciphered in the 19th century contained tables for basic arithmetic operations, cubes, and square roots.
- **Numerical System:** Used only two symbols (1 and 10) combined to form all other numbers.
- **Calculation Tools:** Counting boards (made of wood with grooves for placing beads) existed to assist with counting and simple calculations, similar to an abacus. An abacus differed by having beads with holes placed on specific rods.

The Egyptians

- **Practical Mathematics:** Egyptian mathematics solved practical problems such as measuring time, measuring floods, calculating land area, bookkeeping, and calculating taxes.
- **Numerical System:** Base 10 number system with distinct symbols for 1, 10, 100, 1,000, 10,000, and 100,000.
- **Rhind Papyrus:** A famous writing surface like paper containing examples of arithmetic and geometric problems used to prepare students for the Pharaoh's building program.

The Greeks

- **Timeframe & Core Focus:** Approximately 500-600 BC, centered on Geometry.
- **Pythagoras:** Philosopher and mathematician who studied in Egypt. Founded a secret society (the Pythagoreans) of men and women who believed in the transmigration of souls and that numbers were the essence of all things. Credited with founding the Pythagorean theorem (though Babylonians were likely aware of it 1,000 years prior).

- **Thales:** Best known for Thales' theorem in Euclidean geometry, which states that if A, B, and C are points on a circle where line AC is the diameter, angle $\angle ABC$ is a right angle.

The Romans

- **Influence & Number System:** Influenced by Greek culture and mathematics. Used letters to symbolize numbers in sequences (I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000).
- **Calculation & Modern Use:** Because Roman numerals were difficult to calculate with, abacuses were often used. Modern uses include clocks, building cornerstones, movie credits, and sports events (Olympics and Super Bowl).

The Islamic Influence

- **Origin & Approach:** Originated in North Africa, the Middle East, and Spain as a continuation of Greek achievements.
- **Algebraic Contributions:** Developed a theory using rational and irrational numbers as algebraic objects (moving away from purely geometric concepts). Islamic scholars applied algebra to arithmetic and geometry and studied curves using equations.
- **Significance:** Unprecedented computing power brought exceptional power to science, industry, and business, along with exceptional difficulties and dangers.

Notable Figures in Computing

Wilhelm Gottfried Leibniz (Step Reckoner)

- **Profile:** German philosopher, mathematician, and pioneer in mechanical calculation who developed the binary number system used in digital computers.
- **The Step Reckoner:** Built between 1672–1694 after recognizing that Blaise Pascal's Pascaline could only add and subtract. The Step Reckoner was the first calculator to execute all four basic arithmetic operations (addition, subtraction, multiplication, division) plus the extraction of roots.

Charles Babbage (The Difference Engine)

- **Profile:** Considered one of the grandfathers of computing alongside George Boole. Fascinated by eliminating human error in mathematical tables.
- **Difference Engine:** Designed in 1821 to compute polynomial, logarithmic, and trigonometric functions (like sine and cosine).
- **Ada Lovelace:** Analyst working on the engine, considered by many as the "first computer programmer."
- **Execution:** Babbage only produced prototypes; Swedish engineers George and Edward Schuetz built the first working Difference Engine in 1853.

George Boole (Symbolic Logic)

- **Profile:** Grandfather of computing who developed Boolean algebra, providing the theoretical mathematical model for switching theory and digital circuit design.

- **Key Work:** Published *Mathematical Analysis of Logic* (1847), arguing logic should be a separate branch of mathematics.
- **Logic System:** Introduced 1 (the universe of thinkable objects) and 0 (the absence of things), used symbols (x, y, z) to represent classes of objects, and used three operators ($+$, $-$, $\times$) to combine classes.

Grace Brewster Murray Hopper (COBOL)

- **Profile:** Computer pioneer, naval officer, and Yale graduate.
- **Contributions:** Trailblazing role in developing computer languages, particularly COBOL (Common Business-oriented Language) in 1959. Promoted its adoption in the military and private sectors, making it the most extensively used language in the world by the 1970s.

Katherine Johnson (Human Computer)

- **Profile:** NASA "human computer" who performed complex mathematical calculations for space travel before digital computers existed.
- **Contributions:** Studied space travel geometry to calculate trajectories, Earth orbits, and moon landings for spacecraft, enabling NASA to send astronauts to the moon and back in 1962.

Gladys Mae West (GPS Technology)

- **Profile:** Second Black woman hired at the Naval Proving Ground in Virginia (1956), specializing in large-scale computer and data-processing systems for satellite data analysis.
- **Contributions:** Created a detailed geodetic model of Earth and programmed the computer that calculated Earth's geoid to adequate precision, forming the foundation for Global Positioning System (GPS) development in the 1960s.